

CURRICULUM VITAE

Mark Jarzynski

Graphics Engineer, Oxide Games
Ph.D. Candidate, UMBC

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Education

- 2020 - present University of Maryland, Baltimore County; Ph.D. Student, Computer Science
Admitted to Candidacy: July 2024.
Dissertation: Hashing Out Performance: Large Evaluating and Optimizing Hash Functions for GPU Rendering
- 2007 - 2011 University of Maryland, Baltimore County; B.S., Computer Science

Experience in Higher Education

- 2014 - 2021 University of Maryland, Baltimore County; Imaging Research Center;
Technical Director. All previous duties at IRC plus led and supervised technical teams of students and staff on all IRC technical projects.
- 2011 - 2014 University of Maryland, Baltimore County; Imaging Research Center;
Systems Administrator & Programmer. Maintain IRC's desktops and servers. Developed and maintained mission critical software to meet research needs. Work with students to ensure IRC researcher needs were being met.
- 2010 - 2011 University of Maryland, Baltimore County; Computer Science & Electrical Engineering; UNIX Administrator & Web Developer. Maintained CSEE linux desktops and servers across classroom labs and TA rooms. Designed and developed CSEE's website.

Experience in other than Higher Education

- 2021 - present Oxide Games; Graphics Engineer
2008 - 2010 Zeroed-In Technologies LLC; Programmer

Publications

Dan Baker and Mark Jarzynski, Generalized Decoupled and Object Space Shading System, *Eurographics Symposium on Rendering (ESGR)*, 2022

DOI: <https://doi.org/10.2312/sr.20221163>

Mark Jarzynski and Marc Olano, Hash Functions for GPU Rendering, *Journal of Computer Graphics Techniques (JCGT)*, vol. 9, no. 3, 21-38, 2020

Available online <https://jcg.org/published/0009/03/02/>

Mark Murnane, Don Engel, Stephen Freeland, Lee Boot, Mark Jarzynski, Katrine Lindvig, Line Hillersdal, and David Earle. 2020. Extending CoNavigator into a Collaborative Digital Space. *In Companion of the 2020 ACM International Conference on Supporting Group Work (GROUP '20)*. Association for Computing Machinery, New York, NY, USA, 127–130.

DOI: <https://doi.org/10.1145/3323994.3369890>

Video Games

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| 2026 | ASHES OF THE SINGULARITY II. Real-time strategy game for PC built on the Nitrous 3.0 engine. To be released 2026.
https://www.ashesofthesingularity.com/ |
| 2025 | SINS OF A SOLAR EMPIRE II: PATHS TO POWER. DLC. Real-time 4X space strategy game. Released March 2025.
https://www.sinsofasolarempire2.com/pathstopower |
| 2024 | ARA: HISTORY UNTOLD. 4X historical grand strategy game for PC built on the Nitrous 2.0 engine.. Released September 2024.
https://www.arahistoryuntold.com/ |

Research Projects

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| 2026 - present | HASH FUNCTIONS VIA MINIMIZATION OVER GPU EXECUTION SPEED AND HASH QUALITY. Optimizing hash functions by using an minimization algorithm over gpu execution speed and hash quality, using the test developed in Structure In Noise. |
| 2025 - 2026 | STRUCTURE IN NOISE. Test for evaluating the visual quality of hash functions by searching for matching bit-plane subregions.
https://github.com/markjarzynski/StructureInNoise |
| 2024 - 2025 | FINDING STRUCTURE IN NOISE. Using image feature detectors to find repeating patterns and artifacts in PRNG and hash function images.
https://github.com/markjarzynski/StructureInNoise/tree/feature-detector
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2021	ROBOT TELEOPERATION WITH VIRTUAL REALITY. Study to determine operator efficiency with teleoperating a robot via VR versus traditional 2D screen and controller. https://github.com/markjarzynski/VRTeleoperation
2021	RAY TRACING FOR ACCELERATED METROLOGY. Using Nvidia Optix ray tracing to simulate electron microscopy. https://github.com/markjarzynski/RTAM/tree/develop
2021	EGIS IMPORTER. Plugin for Unreal Engine 4 for importing, manipulating, and displaying geospatial data. https://github.com/ircumbc/EGISImporter
2020 - 2021	LOSING WINTER. AR mobile app to experience and share vlogs of memories of snow and the winter. https://www.losingwinter.net
2020 - 2021	VR COVID-19 DATA VISUALIZATION. VR experience to visualize 3D charts and graphs of various factors of COVID-19 related health data.
2018 - 2020	HASH FUNCTIONS FOR GPU RENDERING. Evaluated existing hash functions for random number quality using TestU01 and GPU execution speed through benchmarking. Published in Journal of Computer Graphics Techniques Oct 17th, 2020. http://icgt.org/published/0009/03/02/ https://www.shadertoy.com/view/XlGcRh https://www.youtube.com/watch?v=dwi5qJ6oPiE&t=65s
2018	UMBC 2018 FACILITIES MASTER PLAN. VR experience of future plans of the UMBC Campus. https://irc.umbc.edu/umbc-2018-facilities-master-plan/
2016 - 2018	ECOMIMESIS. VR experience of 3D photogrammetry scanned plants. Exhibited at Hustle at the Science Lab Gallery Detroit. https://irc.umbc.edu/plantelligence-and-ecomimesis/
2015 - 2019	MAPTU. Cloud based software for mapping and modeling ideas in 3D spaces.
2014 - 2017	RETRIEVER STORIES. Online space for alumni and other members of the UMBC community to share their memories, reconnect with one another, and tell their stories of grit and greatness through the years. https://retrieverstories.umbc.edu/
2014 - 2016	KENDRA. iPad app that allows the user to control a digital crab puppet. Crab may be controlled via touchscreen to walk, move its arms, claws, and eye stalks with inverse kinematics. The crab may also pick up objects throughout

	the space and attach them to its back. Exhibited at Light City Baltimore 2016. https://irc.umbc.edu/hoarder-crab-digital-puppet/
2014 - 2016	BIO MAPPING. Surveyed subjects regarding their health and location. Anonymized health data and fuzzy mapped the information.
2013 - 2016	CRANES IN MOTION/MIMICKING WHOOPER. Interactive 3D application utilizing Microsoft Kinect for motion controlling a 3D whooping crane. On permanent display at the Patuxent National Wildlife Visitor Center. https://irc.umbc.edu/cranes-in-motion/
2013	THREE.SIXTY PANORAMA ENGINE. Web based three hundred and sixty degree panorama viewer. https://irc.umbc.edu/three-sixty-panorama-engine/
2013 - 2014	SHERMAN'S MARCH AND AMERICA: MAPPING MEMORY. Digital map that depicts Sherman's March to the sea during the Civil War. http://shermansmarch.org/
2012 - 2016	BEARINGS OF BALTIMORE, CIRCA 1815. Interactive touch screen display of a digitally recreated 3D map of Baltimore, Maryland circa 1815. On permanent display at the Maryland Historical Society. https://earlybaltimore.org/
2012 - 2015	NATIONAL ACADEMY OF SCIENCES GREAT HALL. iPad app that explores the history of science through an augmented reality tour of the Great Hall ceiling at the National Academy of Sciences building in Washington, D.C. On permanent display at the National Academy of Sciences. https://irc.umbc.edu/dome-explorer/
2012 - 2014	BALTIMORE ART+JUSTICE PROJECT. Social network that brings together individuals and organizations working at the intersection of art and social justice in Baltimore, Maryland. https://irc.umbc.edu/baltimore-artjustice/
2012 - 2014	PARISIENNES: WOMEN AND THE CITY IN FRENCH CINEMA. Research tool used to map scenes from historical French films. https://irc.umbc.edu/parisiennes-women-and-the-city-in-french-cinema/
2011 - 2013	SYMPHONY INTERACTIVE. iPad app which provides contextual notes and additional information during live symphony performances. https://irc.umbc.edu/symphony-interactive/

2011 - 2013	MAPPING BAYBROOK. Digital map illustrating the history and culture of an industrialized community in Baltimore, Maryland. https://irc.umbc.edu/mapping-baybrook/
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Awards

2006	Eagle Scout. Boy Scouts of America. Troop 447.
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Service to the Community

2022 - present	Vice President. Windy Knolls West Homeowners Association.
2008 - 2022	Merit Badge Counselor. Boy Scouts of America. Troop 447.
2006 - 2008	Assistant Scout Master. Boy Scouts of America. Troop 447.